8.4 Spiralling onto a Circle

WORKSHEET

The question

STEP III 2015, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q8

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Both equations contain the combinations $x\,\mathrm{d}x + y\,\mathrm{d}y$ and $x\,\mathrm{d}y - y\,\mathrm{d}x$ in disguise. What are these in polar coordinates?

STAGE 1 Part (i) in polar coordinates

Covers: Part (i)

Ask yourself Rearrange the equation so that it is built from $x\,\mathrm{d}x + y\,\mathrm{d}y$ and $x\,\mathrm{d}y - y\,\mathrm{d}x$. Expand to check.

Pause and try Replace these two combinations by $r\,\mathrm{d}r$ and $r^2\,\mathrm{d}\theta$. What differential equation for r in terms of θ do you get? Solve it.

STAGE 2 Part (ii)

Covers: Part (ii)

Ask yourself The new equation is the old one plus terms containing $x^2 + y^2 = r^2$. Group the new terms: which of the two combinations appears?

Pause and try You should get $\frac{dr}{d\theta} = r^3 - r$. Put $u = r^{-2}$. What linear equation does u satisfy?

STAGE 3 The three kinds of solution

Covers: the shapes of the solutions

Ask yourself Look at $r^2 = \frac{1}{1+Ae^{2\theta}}$ separately for $A = 0$, $A > 0$ and $A < 0$. In each case, what happens to r as $\theta \to -\infty$ and as θ increases? When $A < 0$, which values of θ are allowed?

What's really going on

Limit cycles

Henri Poincaré introduced limit cycles in the 1880s while studying the qualitative behaviour of differential equations in the plane. Interpret the question's equation as describing the path of a point moving in the plane. The unit circle is an isolated closed orbit, and every nearby orbit winds towards it (or away from it, depending on the direction of time). In 1926 the Dutch engineer Balthasar van der Pol found a limit cycle in an electrical circuit with a valve, and proposed that similar equations describe the beating of the heart. Limit cycles model any system with a self-sustained, stable rhythm: circadian clocks, nerve impulses, the cycles of predator and prey populations.

No chaos in the plane

The *Poincaré–Bendixson theorem* says that for a smooth system in the plane, an orbit that stays in a bounded region without approaching an equilibrium must approach a closed orbit. So in two dimensions, long-term behaviour is limited to equilibria and cycles: there is no room for chaos, because orbits cannot cross. Chaos needs at least three dimensions, as Edward Lorenz discovered in 1963 with his famous three-variable model of convection (the "butterfly"). In discrete time, as we will see in Section 8.7, chaos is possible even in one dimension.

An unsolved problem

In 1900 David Hilbert asked, as part of his sixteenth problem: what is the largest number of limit cycles that a system $x' = P(x, y)$, $y' = Q(x, y)$ can have, when P and Q are polynomials of degree n? It is still unknown even for $n = 2$, where systems with

four limit cycles are known and it is not known whether more are possible. It is one of the very few of Hilbert's problems that remain completely open.

Try this next

1 **The spiral.** For the solution $r = e^{-\theta}$ of part (i), show that the curve crosses every ray from the origin at the same angle, 45°. (This is why logarithmic spirals are called *equiangular*.)

2 **Polar identities.** Prove that $x\,\mathrm{d}x + y\,\mathrm{d}y = r\,\mathrm{d}r$ and $x\,\mathrm{d}y - y\,\mathrm{d}x = r^2\,\mathrm{d}\theta$.

3 **A system in time.** Consider the system $\dot{r} = r(1 - r^2)$, $\dot{\theta} = 1$. Show that every solution with $r(0) > 0$ approaches the unit circle as $t \to \infty$. Write the system in terms of x and y.

4 **Two cycles.** Find a system of the form $\dot{r} = f(r)$, $\dot{\theta} = 1$ with two limit cycles, at $r = 1$ and $r = 2$, one attracting and one repelling.

Further reading

- **Steven Strogatz, *Nonlinear Dynamics and Chaos*** (CRC Press). Chapter 7 is about limit cycles and the Poincaré–Bendixson theorem.
- **Steven Strogatz, *Sync*** (Hyperion). A popular book about rhythms and synchronisation in nature.
- **James Gleick, *Chaos: Making a New Science*** (Penguin). The story of Lorenz and the discovery of chaos.

The full guided solution is in Section 8.4:
https://beyondstep.keithmcnulty.org/ch08/limit-cycle.html